II-Probabilistic Approaches Probability, Marginal and Conditional Distributions

Foundations of Neuro-Symbolic AI

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Outline

- Uncertainties
- Probability Distributions
- Marginal Distributions
- Conditional Distributions
- Bayes Theorem
- Connections with logics

Toy accounting example

Let us imagine that each possible assignment to is of equal frequency.

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \quad [X_{A1}, X_{A2}, X_F]$$

We can ask the following questions: Is the account Account 1 or the account Account 2 more likely to be booked?

Since logic can only handle certainty, it cannot answer this question.

However, we can intuitively assign probabilities to the states:

$$\begin{pmatrix} 0 & \frac{1}{3} \\ \frac{1}{3} & 0 \end{pmatrix} \quad \begin{pmatrix} 0 & 0 \\ \frac{1}{3} & 0 \end{pmatrix} \quad [X_{A1}, X_{A2}, X_F]$$

There are two worlds where Account 1 is booked, and one world where Account 2 is booked, thus

$$\mathbb{P}[X_{A1} = 1] = \frac{2}{3} \quad > \quad \mathbb{P}[X_{A2} = 1] = \frac{1}{3}.$$

Uncertainty: Summarizing the unknown

Uncertainties need to be handled:

- Not all variables are measured.
  Example: When we do not know the value of Feature, then we do not know whether Account 1 or Account 2 have to be booked.
- Parts of the models are not known.
  Example: Even if Feature is known to be False, then we do not know whether Account 1 or Account 2 have to be booked.

Logics cannot handle uncertainties.
But: They are already built in model counts.

Comparing logics and probability theory

Modelling assumptions

- **Ontological**: What properties does a system have?
  $\rightarrow$ We assume fixed sets of categorical properties.
- **Epistemological**: What can we tell about these properties?
  $\rightarrow$ We associate possibilities or probabilities with states.

Two main frameworks differing in the empistemological assumptions:

- Logics (Possibilities): The traditional mathematical model of intelligence.
- Probability Theory (Probabilities): Handling uncertainties about the systems state.

In both cases, we encode our knowledge in an arrangement of numbers: Tensors with efficient representations by Tensor Networks.

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Recap: Tensors as Propositional Formulas

So far: Tensors with Boolean coordinates and leg dimension 2
Generalize to

- arbitrary leg dimensions
- real valued coordinates instead of Booleans $\{0, 1\}$

Probability distributions

Instead of storing 0/1 for possibilities, we can store probabilities (in the interval $[0, 1]$) in the coordinates of a tensor.
The total probability of all states must be 1 and can be calculated by contracting all legs of the tensor since

$$\sum_{x_{[d]}} \mathbb{P}\left[X_{[d]} = x_{[d]}\right] = \left\langle \mathbb{P}\left[X_{[d]}\right] \right\rangle_{[\varnothing]} .$$

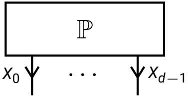
Definition (Probability Tensor)

Let there be a factored system $X_{[d]}$ defined by variables X_k taking values in $[m_k]$. A probability distribution over the states of $X_{[d]}$ is a tensor $\mathbb{P}\left[X_{[d]}\right]$ with coordinates in the interval $[0, 1]$ such that

$$\left\langle \mathbb{P}\left[X_{[d]}\right] \right\rangle_{[\varnothing]} = 1 .$$

Normalization constraint by directionality

Probability distributions sum to 1.
We depict this by outgoing legs:



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Example: Being at a dentist

We are reasoning about a factored system with three variables:

- Toothache $x_0 \in [2]$, whether your tooth hurts
- Cavity $x_1 \in [2]$, whether there is a cavity in your tooth
- Catch $x_2 \in [2]$, whether the dentist catches in your tooth

Example: Being at a dentist

Let us take the following probability distribution over the states of the system:

Toothache x_0	Cavity x_1	Catch x_2	$\mathbb{P}[X_0 = x_0, X_1 = x_1, X_2 = x_2]$
0	0	0	0.576
0	0	1	0.144
0	1	0	0.008
0	1	1	0.072
1	0	0	0.064
1	0	1	0.016
1	1	0	0.012
1	1	1	0.108

We align the probability values in the coordinates of an order-3 tensor:

$$\begin{pmatrix} 0.576 & 0.008 \\ 0.064 & 0.012 \end{pmatrix}\begin{pmatrix} 0.144 & 0.072 \\ 0.016 & 0.108 \end{pmatrix} [X_0, X_1, X_2]$$

Marginal Distributions

Example: What is the probability that there is a cavity (or not) when having toothache (or not)?

- The variable Catch is not relevant for this question, thus we want to marginalize over it.
- To this end, we sum the probability of states differing in the value of Catch only.

This amounts to a contraction of the probability distribution tensor closing the variable X_2:

$$\begin{pmatrix} 0.576 & 0.008 \\ 0.064 & 0.012 \end{pmatrix} \begin{pmatrix} 0.144 & 0.072 \\ 0.016 & 0.108 \end{pmatrix} \quad [X_0, X_1, X_2]$$

$$\downarrow$$

$$\begin{pmatrix} * & * \\ * & * \end{pmatrix} \qquad [X_0, X_1]$$

Marginal Distributions

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$$\downarrow$$

$$\begin{pmatrix} 0.72 & * \\ * & * \end{pmatrix} \quad [X_0, X_1]$$

Marginal Distributions

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$$\downarrow$$

$$\begin{pmatrix} 0.72 & 0.08 \\ 0.08 & 0.12 \end{pmatrix} \; [X_0, X_1]$$

Marginal Distributions

Summation of probabilities of states are done by contractions of the probability distribution tensor.

Definition (Marginal Distribution)

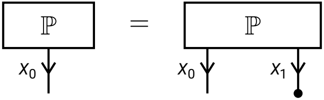
Given a probability distribution $\mathbb{P}\left[X_{[d]}\right]$ and a subset $\mathcal{U} \subset [d]$, the marginal distribution of the variables $X_{\mathcal{U}}$ is the distribution

$$\mathbb{P}\left[X_{\mathcal{U}}\right] = \left\langle \mathbb{P}\left[X_{[d]}\right] \right\rangle_{[X_{\mathcal{U}}]} .$$

Marginal Distributions in Contraction Formalism

Contractions are useful to

- Specify the Probability Tensor, or a decomposition of it
- Specify the variables to marginalize over as the ones left open



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Example: Conditional Distributions

What is the probability of having a cavity when having a toothache? We separately normalize the probabilities of states

$$\begin{pmatrix} 0.72 & 0.08 \\ 0.08 & 0.12 \end{pmatrix}[X_0, X_1] \quad \longrightarrow \quad \begin{pmatrix} * & * \\ * & * \end{pmatrix} \; [X_0|X_1]$$

This is called a conditional distribution.

Example: Conditional Distributions

What is the probability of having a cavity when having a toothache? We separately normalize the probabilities of states

$$\begin{pmatrix} 0.72 & 0.08 \\ 0.08 & 0.12 \end{pmatrix}[X_0, X_1] \quad \longrightarrow \quad \begin{pmatrix} \frac{0.72}{0.72+0.08} & \frac{0.08}{0.72+0.08} \\ * & * \end{pmatrix}[X_0|X_1]$$

This is called a conditional distribution.

Example: Conditional Distributions

What is the probability of having a cavity when having a toothache? We separately normalize the probabilities of states

$$\begin{pmatrix} 0.72 & 0.08 \\ 0.08 & 0.12 \end{pmatrix} [X_0, X_1] \quad \longrightarrow \qquad \begin{pmatrix} 0.9 & 0.1 \\ 0.4 & 0.6 \end{pmatrix} [X_0 | X_1]$$

This is called a conditional distribution.

Normations

More general, conditionalization amounts to normalization of probability tensors.

Definition (Normalization)

The normalization of a tensor $\tau\left[X_{[d]}\right]$ with respect to disjoint subsets $\mathcal{V}^{\mathrm{out}}, \mathcal{V}^{\mathrm{in}} \subset [d]$ is the tensor

$$\left\langle \tau\left[X_{[d]}\right]\right\rangle_{\left[X_{\mathcal{V}^{\mathrm{out}}}|X_{\mathcal{V}^{\mathrm{in}}}\right]}$$

defined by the coordinates

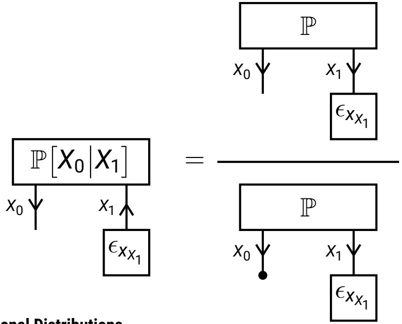
$$\begin{aligned}&\left\langle \tau\left[X_{[d]}\right]\right\rangle_{\left[X_{\mathcal{V}^{\mathrm{out}}}=x_{\mathcal{V}^{\mathrm{out}}}|X_{\mathcal{V}^{\mathrm{in}}}=x_{\mathcal{V}^{\mathrm{in}}}\right]}\\ &\quad=\begin{cases}\dfrac{\left\langle \tau\left[X_{[d]}\right]\right\rangle_{\left[X_{\mathcal{V}^{\mathrm{out}}}=x_{\mathcal{V}^{\mathrm{out}}},X_{\mathcal{V}^{\mathrm{in}}}=x_{\mathcal{V}^{\mathrm{in}}}\right]}}{\left\langle \tau\left[X_{[d]}\right]\right\rangle_{\left[X_{\mathcal{V}^{\mathrm{in}}}=x_{\mathcal{V}^{\mathrm{in}}}\right]}} & \text{if} \quad \left\langle \tau\left[X_{[d]}\right]\right\rangle_{\left[X_{\mathcal{V}^{\mathrm{in}}}=x_{\mathcal{V}^{\mathrm{in}}}\right]} \neq 0\\ \dfrac{1}{\prod_{v\in\mathcal{V}^{\mathrm{out}}} m_v} & \text{else}\end{cases}\end{aligned}$$

Formal Definition of Conditional Distributions

Definition (Conditional Probability)

Given a distribution $\mathbb{P}\left[X_{[d]}\right]$ and disjoint subsets $\mathcal{V}^{\mathrm{out}}, \mathcal{V}^{\mathrm{in}} \subset [d]$, the distribution of $X_{\mathcal{V}^{\mathrm{out}}}$ conditioned on $X_{\mathcal{V}^{\mathrm{in}}}$ is the normalization

$$\mathbb{P}\left[X_{\mathcal{V}^{\mathrm{out}}} \middle| X_{\mathcal{V}^{\mathrm{in}}}\right] = \left\langle \mathbb{P}\left[X_{[d]}\right] \right\rangle_{\left[X_{\mathcal{V}^{\mathrm{out}}} | X_{\mathcal{V}^{\mathrm{in}}}\right]}$$



Directed notation

Directed tensors represent normations:
Each slice of the incoming variables is a normalized tensor, i.e. coordinates summing to 1.

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The Bayes Theorem

The Bayes Theorem relates conditional probabilities:

Theorem (Bayes Theorem)

For any joint distribution of two categorical variables X_0 and X_1 it holds that

$$\mathbb{P}\left[X_0 \middle| X_1\right] = \frac{\mathbb{P}\left[X_0, X_1\right]}{\mathbb{P}\left[X_1\right]} = \frac{\mathbb{P}\left[X_1 \middle| X_0\right]\mathbb{P}\left[X_0\right]}{\mathbb{P}\left[X_1\right]} .$$

Bayes Theorem in the Dentist Example

Directions of Reasoning

- Causal direction: Toothache is caused by cavity
- Diagnostic direction: Cavity is probable because of toothache

Bayes Theorem allows us to reason in diagnostic direction, given an underlying causal influence:

$$\begin{aligned}&\mathbb{P}[\text{Cavity}|\text{Toothache}]\\&\qquad= \mathbb{P}[\text{Toothache}|\text{Cavity}]\frac{\mathbb{P}[\text{Cavity}]}{\mathbb{P}[\text{Toothache}]}\end{aligned}$$

Contraction Calculus for Probability Tensors

Probabilistic queries can be answered by contractions

- Marginal probabilities

$$\mathbb{P}[X_0] = \langle \mathbb{P} \rangle_{[X_0]}$$

- Conditional probabilities

$$\mathbb{P}\left[X_0 \middle| X_1\right] = \langle \mathbb{P} \rangle_{[X_0|X_1]}$$

Outlook: Tensor network decompositions of $\mathbb{P}$ increase the execution efficiency!

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Probabilistic Queries

The probability of a formula f to hold

$$\sum_{x_{[d]}\,:\,f\left[X_{[d]}=x_{[d]}\right]=1} \mathbb{P}\left[X_{[d]}=x_{[d]}\right] = \left\langle \mathbb{P}\left[X_{[d]}\right], f\left[X_{[d]}\right] \right\rangle_{[\varnothing]}$$

Generalization of entailment: A formula is entailed if the probability is 1.